

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.9 LINEAR MOTION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we use mathematics, through calculations and graphs, to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway? In the 2023 Nairobi Marathon, Runner A (nearly constant pace) won, even though Runner B (fast start, slow middle, sprint finish) was ahead at 30 km. Use everything you learned in Lessons 1 to 8 to answer the five sections below. Write in full sentences, show every step of your working, include units, and refer back to the marathon in your answers. Where a sketch is asked for, draw it neatly with labelled axes. Use your class notes, graphs and Driving Question Board to help you.

Section 1: The language of motion (Key Inquiry Question 1: What is speed?)

(a) Explain the difference between distance and displacement, and between speed and velocity. State which of these are scalar quantities and which are vector quantities. **(b)** Both runners covered 42 km in the Nairobi Marathon. Explain why comparing only their total times could not tell you that Runner B was ahead at 30 km. **(c)** Define acceleration and say which part of Runner B's race it helps us describe.

Write your answer here:

Section 2: Calculating velocity and acceleration

(a) Runner A ran at a nearly constant velocity of about 19 km/h. Use $v = s/t$ to find how far Runner A runs in 2 hours. (b) Runner B's velocity in the middle of the race was about 17.6 km/h. Calculate how much further Runner A runs than Runner B in 30 minutes (0.5 h) at these velocities. (c) A boda-boda on Mombasa Road accelerates from rest at 1 m/s^2 . Use $a = \Delta v/\Delta t$ to find its velocity after 10 seconds. (d) Is Runner A's motion uniform or non-uniform? What is his acceleration? What does a negative acceleration mean for Runner B?

Write your answer here:

Section 3: Displacement-time graphs (Key Inquiry Question 2: How do we use displacement-time graphs in real life?)

(a) Say what each of these shapes on a displacement-time graph tells you about the motion: a steep straight line, a gentle straight line, a horizontal line, a line sloping back down. (b) State what the gradient of a displacement-time graph measures. (c) Describe in words how the lines for Runner A and Runner B would look on the same displacement-time graph, from the start to the finish. Say where the lines cross and what the crossing shows. (d) Give one real-life Kenyan example where a displacement-time graph is useful, for example a matatu on the Nakuru highway.

Write your answer here:

Section 4: Velocity-time graphs (Key Inquiry Question 3: How do we use velocity-time graphs?)

(a) State what the gradient of a velocity-time graph measures. State what the area between the graph and the time axis measures. (b) Sketch a velocity-time graph for a matatu on the Nakuru–Nairobi highway that accelerates from a stage, cruises at constant velocity, then brakes. Label each of the three sections as acceleration, constant velocity or deceleration. (c) For the 30-minute part of the race in Section 2(b), the velocity-time graph for each runner is a horizontal line. Use area under the graph to find the distance each runner covers, and say whose area is larger and what that means for the race.

Write your answer here:

Section 5: Relative speed and the full story of the marathon

(a) Runner B had slowed to about 17.6 km/h and Runner A was running at about 19 km/h. Calculate the relative speed of Runner A with respect to Runner B. Then find how many minutes Runner A needs to close a gap of 200 m (0.2 km) to Runner B. (b) A matatu travelling at 90 km/h overtakes a bus travelling at 60 km/h in the same direction. Find the relative speed. (c) Two vehicles each travelling at 80 km/h approach each other in opposite directions. Find the relative speed, and explain why this differs from part (b). (d) Now write a final explanation of the driving question. Use the Nairobi Marathon to explain why Runner A won even though Runner B was ahead at 30 km. In your answer refer to average speed and instantaneous velocity, acceleration, the shape of a displacement-time graph, and a velocity-time graph.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of key terms (distance/displacement, speed/velocity, scalar/vector, acceleration)	Defines all terms correctly, correctly labels each as scalar or vector, and uses them accurately and consistently throughout.	Defines most terms correctly with only minor slips in labelling scalar or vector.	Confuses distance with displacement or speed with velocity, or omits scalar/vector.
Calculations ($v = s/t$, $a = \Delta v/\Delta t$, area under graph, relative speed)	All calculations are correct with clear working, correct formulas and correct units.	Most calculations are correct; occasional errors in units or working.	Several incorrect calculations or formulas, little working or missing units.
Graph interpretation (gradient and area)	Correctly states that the gradient of a displacement-time graph is velocity, the gradient of a velocity-time graph is acceleration and the area is	States the meaning of gradient and area correctly, but sketches or links to the runners are partly incomplete.	Mixes up what the gradient or area means on the two types of graph, or sketches are unlabelled.

	displacement. Sketches are neat, labelled, and linked to Runner A and Runner B.		
Explaining the phenomenon with evidence	Gives a complete, logical explanation of why Runner A won despite Runner B leading at 30 km, using velocity, acceleration, graph shape and relative speed.	Explains the main reason (B slowed in the middle) using at least two ideas from the unit.	Only restates that A won, or uses average speed alone, with little evidence.
Kenyan contexts and communication	Links ideas clearly to Kenyan examples (matatu, boda-boda, Nairobi Marathon) in clear, well-organised sentences.	Uses some Kenyan examples; writing is mostly clear.	Few or no links to the contexts; explanations are unclear or very short.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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